

Weekly Notice 12

Exercises, **May 1:**

1. Durrett, Exercise 2.3.
2. Durrett, Exercise 2.18. Also determine the probability of collision if time to pass the road is uniform on the interval $(2, 5)$ (where the time unit is seconds).
3. Durrett, Exercise 2.19.
4. Durrett, Exercise 2.36.
5. Exercise 113.
6. Exercise 132. The filtration $(\overline{\mathcal{F}}_t)_{t \geq 0}$ is defined just before Corollary 9.19.
7. Let $X_i \sim \text{Exp}(\lambda_i)$ for $i = 1, 2$, where $\lambda_i > 0$, and assume that X_1 and X_2 are independent.
 - (a) Determine the distribution of $X_1 \wedge X_2$.
 - (b) Determine the probability $\mathbb{P}(X_i = X_1 \wedge X_2)$ (i.e., the probability that X_i is the smallest of X_1 and X_2) for $i = 1, 2$.